

Fashion Recommendation System

Using Collaborative Filtering and Cosine Similarity

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Phase 4 Project – Advanced

Business Problem

- Fashion platforms contain thousands of products, making product discovery difficult for users.
- Generic recommendations reduce user engagement and personalization.
- Businesses need recommendation systems to improve customer experience and increase sales.

Build a recommendation system that generates personalized fashion product recommendations based on user rating behavior.

Project Objective

- Analyze fashion product and user rating data
- Build recommendation models for personalized suggestions
- Implement collaborative filtering using cosine similarity
- Compare baseline and advanced recommendation approaches
- Visualize insights using Tableau dashboards
- Deploy recommendations through a Streamlit application

Dataset Overview

Feature	Description
User ID	Unique user identifier
Product ID	Unique product identifier
Product Name	Fashion product
Brand	Product brand
Category	Product category
Rating	User rating
Color	Product color
Size	Product size
Price	Product price

Dataset contains user-product interactions used to generate personalized recommendations.

Data Preprocessing

- Encoded categorical variables using LabelEncoder
- Created a user-item interaction matrix
- Handled missing values using zero imputation
- Built a preprocessing pipeline for workflow consistency
- Prepared data for similarity computation

Baseline vs Advanced Model

Baseline Model

Popularity-Based Recommendation

- Recommends highest-rated products overall
- Simple and fast
- Works for new users
- Not personalized

Advanced Model

Collaborative Filtering

- Uses cosine similarity
- Identifies similar users
- Generates personalized recommendations
- More relevant recommendations

Recommendation Workflow

User Ratings



User-Item Matrix



Cosine Similarity



Similar Users



Personalized Recommendations

The system identifies users with similar rating patterns and recommends products highly rated by similar users.

Evaluation Metrics

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- Recommendation visualizations
- Similarity heatmaps

Lower RMSE and MAE values indicate better recommendation accuracy and prediction performance.

Tableau Dashboard

An interactive Tableau dashboard was created to visualize:

- Distribution of products by category
- Average ratings across brands
- Product rating distributions

The dashboard helps stakeholders understand:

- Product trends
- Customer preferences
- Brand performance

- Kid's Fashion had the highest product count.
- Gucci products showed the highest average ratings.
- Most product ratings were concentrated between 2.5 and 4.5.

Streamlit Web Application

A simple interactive web application was developed where:

- Users enter a User ID
- The system generates personalized recommendations
- Recommended products are displayed instantly

The application demonstrates how the recommendation system works in practice

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Key Insights

- Users with similar rating behaviors tend to prefer similar fashion products.
- Collaborative filtering improved personalization compared to popularity-based recommendations.
- Popularity-based recommendations provided stable fallback suggestions for new users.
- Product ratings were mostly concentrated between moderate and high values.
- Brand rating differences were relatively small across the dataset.

Challenges Encountered

- Sparse user-item interaction matrix
- Cold-start problem for new users
- Limited recommendation diversity
- Synthetic dataset limitations

Conclusion & Future Improvements

Conclusion

Collaborative filtering successfully generated personalized fashion product recommendations using user similarity patterns.

Future improvements could include:

- Hybrid recommendation systems
- Deep learning recommenders
- Real-time recommendations
- Larger real-world datasets

Thank You



Thank You